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FULL RESEARCH PAPER
Biomarkers

Risk-weighted apoB and lipoprotein-associated coronary risk across ethnic groups

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Aims Apolipoprotein B (apoB) reflects the number of circulating atherogenic lipoproteins but does not account for differences in particle atherogenicity. Risk-weighted apoB (RW-apoB), which incorporates triglyceride-rich lipoproteins and lipoprotein(a) (Lp(a)), may better capture lipoprotein-driven coronary risk, but its performance across diverse populations remains uncertain. We assessed whether the pre-specified triglyceride—and Lp(a)-weighted information captured by RW-apoB was associated with coronary events beyond apoB and compared the discrimination of RW-apoB and measured apoB across ethnicities.

Methods and results We analysed 40 132 participants with 4509 coronary events from Atherosclerosis Risk in Communities Study, Coronary Artery Risk Development in Young Adults Study, Multi-Ethnic Study of Atherosclerosis, the Framingham Offspring Study, and China Kadoorie Biobank (CKB). RW-apoB was calculated using a pre-specified equation. Mutually adjusted Cox estimates were pooled within ethnic groups using random-effects meta-analysis. Separate single-marker models compared discrimination. Sensitivity analyses included age and sex adjustment, standardized exposures, hard-events, truncated follow-up, and CKB-specific logistic regression. Mean RW-apoB exceeded apoB by approximately 8 mg/dL in African American and Chinese participants, reflecting higher Lp(a) and triglycerides, respectively. RW-apoB was associated with coronary events conditional on apoB in White, African American, Chinese, and Hispanic participants: hazard ratios per 10 mg/dL were 1.11 (95% confidence interval 1.05–1.18), 1.11 (1.08–1.15), 1.08 (1.05–1.11), and 1.10 (1.01–1.21), respectively. C-indices for RW-apoB vs. apoB were 0.630 vs. 0.617 in White ($P < 0.001$), 0.606 vs. 0.595 in African American ($P = 0.085$), 0.567 vs. 0.554 in Chinese ($P = 0.014$), and 0.540 vs. 0.535 in Hispanic participants ($P = 0.756$).

Conclusion RW-apoB captured lipoprotein-associated risk beyond measured apoB. Discrimination gains were statistically demonstrable in the larger White and Chinese groups.

Lay summary A measure combining apolipoprotein B (apoB) with triglycerides and lipoprotein(a), termed risk-weighted apoB (RW-apoB), provided additional information about coronary heart disease risk beyond apoB alone, although its value for routine clinical risk prediction remains to be established.

Among more than 40 000 participants from diverse populations, differences between RW-apoB and apoB mainly reflected higher lipoprotein(a) in African American participants and higher triglycerides in Chinese participants.

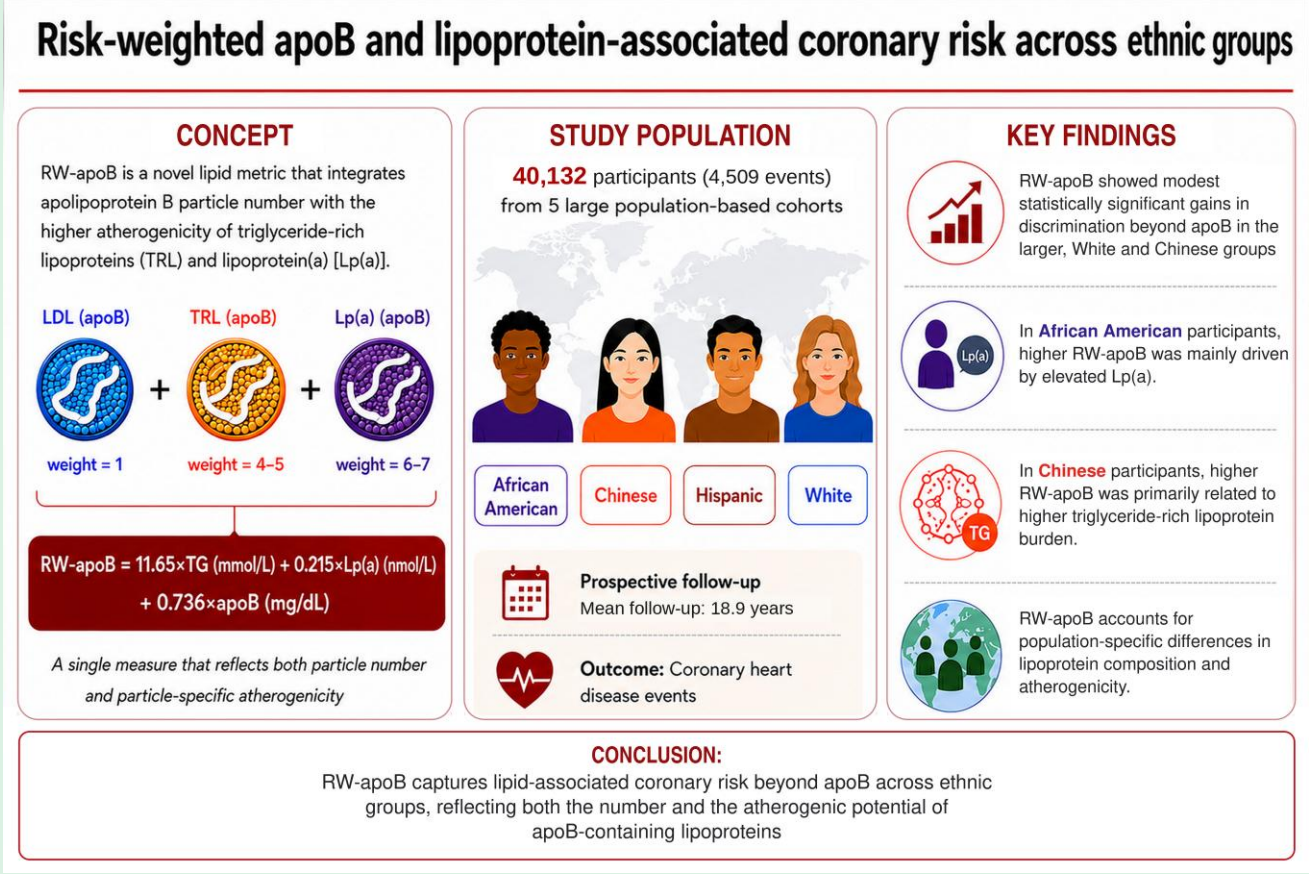
RW-apoB was associated with coronary events after accounting for apoB. Its ability to distinguish participants with earlier events was modestly higher than that of apoB in all four groups, with statistically significant improvements only demonstrable in the larger, White and Chinese groups.

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Graphical Abstract



Keywords Apolipoprotein B • Triglycerides • Lipoprotein(a) • Coronary disease • Ethnicity • Risk stratification

Introduction

Apolipoprotein B (apoB)-containing lipoproteins play a central causal role in atherosclerotic cardiovascular disease (ASCVD). Traditionally, low-density lipoprotein cholesterol (LDL-C) has been the primary therapeutic target, but increasing evidence supports either non-HDL-cholesterol (non-HDL-C) or, preferably, apoB as a more complete marker of atherogenic particle burden, as each atherogenic lipoprotein particle contains one apoB molecule.[1–3] Previous studies have further highlighted the prognostic relevance of apolipoprotein-based measures in cardiovascular disease.[4]

Recent genetic and epidemiological studies have, however, demonstrated that apoB-containing lipoproteins differ in their atherogenic potential. Triglyceride-rich lipoproteins (TRLs) and lipoprotein(a) (Lp(a)) appear to confer greater coronary risk per apoB than LDL, with estimates suggesting approximately 4–5-fold and 6–7-fold higher atherogenicity, respectively.[5–9] These findings suggest that metrics based solely on particle number may incompletely capture true atherogenic burden.

To address this limitation, risk-weighted apoB (RW-apoB) has been proposed as a metric that integrates particle number and particle-specific atherogenicity. The amount of apoB estimated to be present in TRL and Lp(a) is weighted according to published estimates of their relative per-particle atherogenicity compared with LDL. In prior work in the UK Biobank, RW-apoB showed higher discrimination among individual lipid markers and re-ranked a substantial proportion of participants compared with apoB, LDL-C, and non-HDL-C.[10,11]

However, an important and unresolved question is whether the performance and interpretation of RW-apoB are consistent across different ethnic groups. Lipoprotein profiles vary substantially by ethnicity: individuals of African ancestry tend to have higher Lp(a) levels, whereas Chinese populations often exhibit higher triglyceride (TG) levels at comparable apoB concentrations.[12,13] These differences imply that the composition of apoB—and thus the relative contribution of TRL and Lp(a) to overall risk—may differ systematically across populations. Consequently, apoB may underestimate coronary risk to varying degrees depending on the underlying lipoprotein phenotype.

Whether the distribution, interpretation, and prognostic associations of RW-apoB are consistent across diverse populations has not been systematically evaluated. It also remains unclear whether individuals re-ranked by RW-apoB represent similar or distinct lipid phenotypes across ethnic groups and whether RW-apoB contains lipoprotein-associated prognostic information beyond measured apoB within each group.

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In this study, we evaluated the distribution and prognostic performance of RW-apoB compared with apoB across four major ethnic groups (White, African American, Chinese, and Hispanic) in five population-based cohorts. We examined the phenotypic characteristics of individuals re-ranked by RW-apoB and assessed whether the TG- and Lp(a)-weighted information captured by RW-apoB was associated with coronary events beyond measured apoB within each ethnic group.

Methods

Study populations

We assembled data from four major prospective US cohorts—the Atherosclerosis Risk in Communities Study (ARIC), Coronary Artery Risk Development in Young Adults Study (CARDIA), Multi-Ethnic Study of Atherosclerosis (MESA), and Framingham Offspring Study (FOS)—and from the China Kadoorie Biobank (CKB). Data for the US cohorts were obtained from the NHLBI Biologic Specimen and Data Repository Information Coordinating Center. Research protocols were approved by the institutional review board at each participating institution, and all participants provided written informed consent. The present analysis was approved by the ethics committee of Lyon-Villeurbanne Hospital, France. We included participants with available apoB, Lp(a), TG, outcome, and follow-up data. Lipid measurements were obtained from fasting samples in the four US cohorts and from non-fasting samples in CKB. Participants receiving lipid-lowering therapy or with baseline ASCVD were excluded.

RW-apoB was calculated as previously published[11]:

$$11.65 \times TG\,(\text{mmol/L}) + 0.215 \times \text{lipoprotein(a)(nmol/L)} + 0.736 \times \text{apoB (mg/dL)}$$

The coefficients were derived by estimating TRL particle concentration from TGs (assuming that 1 mmol/L of TG corresponds, on average, to 85 nmol/L TRL particles), treating Lp(a) concentration in nmol/L as the Lp(a)-apoB particle concentration, and calculating LDL-apoB by subtraction from total apoB. LDL-, TRL-, and Lp(a)-apoB were then weighted by relative per-particle atherogenicity factors of 1, 4.5, and 6.5, respectively, based on previously published genetic analyses.[4–9,11] The result was converted to the notional apoB unit mg/dL and scaled so that median RW-apoB approximated median measured apoB in the UK Biobank. The same pre-specified equation was applied unchanged in all ethnic groups to evaluate its transportability rather than re-estimating group-specific coefficients.

The ARIC study is a prospective observational cohort that enrolled adults aged 45–64 years from four US communities between 1987 and 1989.[14] The present analysis included 13 240 participants (3314 African American and 9926 White), among whom 2078 incident coronary heart disease (CHD) events occurred during a mean follow-up of 23.0 years, with event ascertainment through 2019. Lp(a) was reported as the measured protein component of the Lp(a) particle, comprising apolipoprotein(a) and apoB, rather than as total particle mass. Following the published ARIC protocol (https://pmc.ncbi.nlm.nih.gov/articles/PMC3760720/#S5), values were multiplied by 3 to approximate total Lp(a) mass and then converted from mg/dL to nmol/L using a factor of 2.2.[15] CHD was defined as the first occurrence of coronary death or nonfatal myocardial infarction (MI).

The CARDIA study is a prospective observational study that enrolled individuals aged 18–30 years between 1985 and 1986, from four cities in the USA (Birmingham, Alabama; Oakland, California; Chicago, Illinois; and Minneapolis, Minnesota).[16] We analysed data from 3610 individuals with 142 CHD events over a mean follow-up of 32.6 years. CHD events were available up to 2022. 1747 individuals self-reported as African American and 1863 as White. Lp(a) was reported in mg/dL and multiplied by 2.2 to obtain a value in nmol/L. CHD events were defined as MI, non-MI acute coronary syndrome, coronary revascularization, and coronary death.

The MESA study is a prospective observational study that enrolled individuals aged 44–84 years between 2000 and 2002.[17] We analysed data from 4434 individuals with 493 events over a mean follow-up of 15 years. CHD events were available up to 2020. 1260 individuals self-reported as African American, 1629 as White, 1007 as Hispanic, and 538 as Chinese. ApoB results were taken from the data field apobd1. Lp(a) was reported in nmol/L. CHD events were defined as MI, resuscitated cardiac arrest, definite angina, probable angina, followed by revascularization, and CHD-related death. The exclusion of participants with apoB <50 mg/dL in MESA followed the approach used in our previous European Heart Journal analysis.[11] Such low apoB concentrations are uncommon without treatment and raised concern about lipid-lowering therapy that was not recorded in the MESA medication data. This concern was further supported by the fact that the 10% event rate in this subgroup was similar to that observed in the whole MESA cohort. This exclusion removed 211 participants with 21 CHD events. Its effect was examined in a sensitivity analysis retaining these participants (see Supplementary material online, *Table S7*).

FOS, a prospective, community-based epidemiological survey, enrolled individuals aged 30–71 years between 1971 and 1975.[18] They were either the biological children or spouses of the children of the original Framingham Heart Study cohort. We analysed data from 2393 individuals with 520 events over a mean follow-up of 38.5 years. CHD events were available up to 2021. All subjects self-reported as White. Lp(a) was reported in mg/dL and multiplied by 2.2 to obtain a value in nmol/L. CHD was defined as MI, angina, coronary revascularization, or CHD death.

CKB is a large prospective cohort study including 512 724 adults aged 30–79 years, recruited from 10 geographically diverse regions in China (five urban and five rural) between 2004 and 2008.[19] At baseline, participants completed a standardized questionnaire capturing sociodemographic characteristics, lifestyle factors, medical history, and medication use, and underwent physical examination. A 10-mL non-fasting blood sample was collected and stored for long-term use. Follow-up for incident disease and cause-specific mortality was conducted through linkage to regional disease and death registries, as well as to health insurance databases capturing hospital admissions. CHD was defined as acute MI. The study was approved by relevant local, national, and international ethics committees, and all participants provided written informed consent. Plasma levels of Lp(a) in 18 174 participants in a nested case–control study of CHD and stroke types in CKB were measured in molar units using a polyclonal assay (Denka Seiken) in the Wolfson Laboratory, Nuffield Department of Population Health, University of Oxford. Here, we included 16 455 individuals with Lp(a), TG, and apoB measurements available, with 1276 events over a mean follow-up of 10.0 years. This was an outcome-dependent biomarker sample, not a random sample of the full CKB cohort; its descriptive event rates and discrimination estimates therefore apply to the analysed sample.

Variables were harmonized to common units and definitions where possible before analysis. ApoB was expressed in mg/dL, TG in mmol/L, and Lp(a) in nmol/L. Cohort-specific assay methods, fasting status, Lp(a) conversions, inclusion criteria, and CHD definitions are described above. Participants with missing apoB, TG, Lp(a),

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outcome, or follow-up data required for the analysis were excluded. The combined analytical sample comprised 40 132 individuals, with mean observed follow-up of 18.9 years and 4509 CHD events. Available exclusion counts are summarized in Supplementary material online, *Table S1*. Observed follow-up extended from baseline to the first CHD event or censoring; descriptive event rates were calculated as events divided by the sum of individual observed person-years, multiplied by 100. The cohorts were included to evaluate whether the relative association of RW-apoB and apoB with CHD was reproducible across heterogeneous populations and study settings; they were not treated as a single homogeneous target population. Outcome associations were therefore estimated within cohort before pooling within ethnic groups.

Statistical analysis

Discordance and re-ranking analyses—Within each ethnic group, apoB and RW-apoB were categorized into ethnicity-specific quintiles. Discordance was described by cross-tabulating apoB quintiles within each RW-apoB quintile and displaying the proportions in stacked bar charts. To characterize the participants most affected by re-ranking, we calculated the ethnicity-specific percentile difference (RW-apoB percentile minus apoB percentile) and divided this difference into quintiles. Q1 represented the greatest downward re-ranking by RW-apoB relative to apoB, whereas Q5 represented the greatest upward re-ranking. Participant characteristics were summarized for Q1 and Q5 within each ethnic group as mean (standard deviation) or *n* (%), as appropriate.

Distribution analyses—Kernel density plots were used to compare the full distributions of apoB, TG, Lp(a), and RW-apoB across ethnic groups. These plots display distribution shape, spread, skewness, and overlap and illustrate how variation in TG and Lp(a) translates into the RW-apoB distribution.

Associations with coronary outcomes—In the total pooled cohort and within each ethnic group, mutually adjusted Cox proportional hazards models including apoB and RW-apoB were fitted separately in each contributing cohort. Hazard ratios (HRs) were estimated per 10 mg/dL higher concentration. Cohort-specific log HRs and standard errors were pooled using random-effects meta-analysis with restricted maximum likelihood estimation in the *metafor* package. Between-cohort heterogeneity was quantified using Cochran's Q statistic, τ^2, and I^2. For an ethnic group represented by a single cohort, the cohort-specific estimate was reported without estimation of between-cohort heterogeneity.

Because RW-apoB = 0.736 × apoB + *W*, where *W* = 11.65 × TG + 0.215 × Lp(a), including apoB and RW-apoB in the same model is a reparameterization of a model containing apoB and this pre-specified weighted TG/Lp(a) component. The conditional RW-apoB coefficient tests whether *W* contributes prognostic information at a fixed apoB concentration. The conditional apoB coefficient instead describes variation in apoB at fixed RW-apoB, which entails compensating variation in *W*; it is not an estimate of the overall association of apoB with CHD. These models address information beyond apoB, not superiority of two independent predictors. The primary models included only the two markers to characterize lipoprotein-associated information; age- and sex-adjusted models were examined separately. Spearman correlations and VIFs from the linear relationship between the markers were calculated within each cohort–ethnicity combination (see Supplementary material online, *Table S2*). Correlation was expected from the formula, and conditional estimates were interpreted cautiously because multicollinearity can inflate uncertainty.

Sensitivity analyses—Mutually adjusted models were repeated with age and sex adjustment and with each marker expressed per cohort-specific standard deviation. Proportional hazards assumptions were assessed using scaled Schoenfeld residuals within each cohort–ethnicity group. Only CARDIA was excluded from this analysis because of the limited number of CHD events (*n* = 142). Follow-up was also truncated at 10 years in ARIC and FOS, censoring events occurring after 10 years at that time. We selected the cohorts with the longest follow-up and enough events to perform this analysis. In MESA, the low-apoB exclusion was reversed in a pooled sensitivity analysis (see Supplementary material online, *Tables S3–S7*). Mutually adjusted models were repeated using hard coronary events only (MI and coronary death, where available) in cohort–ethnicity groups with adequate event numbers.

CKB sampling sensitivity analyses—Cox regression was retained to provide a common time-to-event summary across cohorts, but the CKB models were unweighted and did not account for outcome-dependent sampling. We therefore also fitted CKB-specific unconditional logistic models containing both markers, first without and then with age, age squared, sex, and the 10 study regions. Odds ratios (ORs) were expressed per 10 mg/dL. Single-marker receiver operating characteristic (ROC) areas under the curve (AUCs) were compared using paired DeLong tests (see Supplementary material online, *Table S8*). These analyses assess sensitivity to model specification within the selected sample; they do not correct the original Cox estimates for selection.

Within-cohort discordance and events—Descriptive event rates were calculated across joint, population-specific apoB and RW-apoB tertiles in White ARIC and Chinese CKB participants, and across median categories in African American MESA participants because of fewer events. Rates were expressed per 100 observed person-years. These analyses characterize outcomes within cohorts rather than pooling absolute rates across different populations (see Supplementary material online, *Tables S9–S11*).

Assessment and comparison of discrimination—Discrimination was assessed using Harrell's concordance index (C-index) from separate Cox models for LDL-C, non-HDL-C, apoB, and RW-apoB. Analyses were performed within the total pooled cohort and separately within each ethnic group. When an ethnic group included more than one cohort, models were stratified by cohort to allow cohort-specific baseline hazards while estimating a common predictor effect; unstratified models were used for single-cohort groups.

The C-index and its standard error were obtained from each fitted Cox model, and 95% confidence intervals (CIs) were calculated using the normal approximation. The C-index is the probability that, for a comparable pair, the participant with the higher predicted risk experiences the event earlier.

Differences between the apoB and RW-apoB C-indices were evaluated using a paired Wald test based on the joint variance–covariance matrix returned by the concordance function in the R *survival* package, thereby accounting for estimation in the same participants. Two-sided *P*-values were obtained from the standard normal distribution.

These analyses compare the discrimination of individual lipid markers and were not intended to assess calibration or incremental performance beyond a comprehensive clinical risk model. All analyses were conducted in R using the survival, broom, tidyverse, and metafor packages.

Results

Study population

The study included 40 132 participants from five cohorts, with 4509 CHD events during follow-up: 15 811 White participants (2323 events), 6321 African American participants (738 events), 16 993 Chinese participants (1326 events), and 1007 Hispanic participants (122 events). Baseline characteristics are shown in *Table 1*.

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Table 1 Baseline characteristics of the study population stratified by ethnicity

Characteristic	White *N* = 15 811	African American *N* = 6321	Chinese *N* = 16 993	Hispanic *N* = 1007
Events *n* (%)	2323 (14.7%)	738 (11.7%)	1326 (7.8%)	122 (12.1%)
Follow-up time years (mean)	26.2	22.8	10.2	14.5
Event rate per 100 person-years	0.56	0.51	0.76	0.84
LIPIDS				
apoB mg/dL				
Mean (SD)	94.0 (26.6)	92.0 (27.7)	84.8 (20.8)	98.6 (23.7)
Median (IQR)	91.0 (75.0, 110.0)	89.0 (73.0, 107.4)	84.0 (70.1, 98.1)	96.9 (82.3, 112.0)
apoB 20th percentile	72.0	69.0	66.7	78.5
apoB 40th percentile	85.0	82.7	78.7	92.3
apoB 60th percentile	98.0	96.0	89.4	101.9
apoB 80th percentile	114.6	113.0	101.7	116.5
apoB Range	12.0–294.0	17.0–292.0	20.1–191.5	50.3–225.9
RW-apoB mg/dL				
Mean (SD)	95.0 (28.9)	99.9 (31.6)	93.3 (26.7)	102.9 (28.6)
Median (IQR)	91.3 (74.3, 111.4)	95.2 (78.0, 117.0)	90.0 (74.2, 109.1)	98.5 (82.5, 118.1)
RW-apoB 20th percentile	70.7	73.9	70.5	79.1
RW-apoB 40th percentile	84.6	88.1	83.8	92.0
RW-apoB 60th percentile	98.4	103.2	96.6	106.1
RW-apoB 80th percentile	116.7	122.9	114.1	122.9
RW-apoB Range	16.7–368.7	17.7–324.9	21.3–255.5	45.9–294.5
Triglycerides mmol/L				
Mean (SD)	1.4 (1.0)	1.1 (0.8)	1.9 (1.3)	1.7 (0.9)
Median (IQR)	1.1 (0.8, 1.7)	0.9 (0.7, 1.3)	1.6 (1.1, 2.3)	1.5 (1.1, 2.2)
Lp(a) nmol/L				
Mean (SD)	46.2 (56.2)	89.1 (76.1)	39.8 (47.7)	47.3 (71.2)
Median (IQR)	23.1 (9.8, 60.7)	68.6 (33.8, 122.1)	20.6 (10.5, 48.0)	19.0 (10.0, 49.5)
LDL-C mg/dL mean (SD)	129.9 (36.7)	126.4 (39.7)	93.2 (27.2)	122.6 (31.6)
HDL-C mg/dL mean (SD)	51.2 (16.0)	54.4 (16.1)	47.9 (11.4)	47.1 (12.8)
Non-HDL-C mg/dL mean (SD)	153.8 (43.3)	145.9 (43.8)	133.3 (35.2)	153.0 (35.8)
OTHER CHARACTERISTICS				
Age (years) mean (SD)	49.6 (12.4)	47.1 (15.8)	57.2 (10.5)	61.2 (10.4)
BMI (kg/m^2) mean (SD)	26.9 (4.9)	30.0 (6.3)	23.5 (3.5)	29.4 (5.0)
Male, *n* (%)	7354 (46.8%)	2640 (41.8%)	8537 (50.2%)	499 (49.6%)
Female, *n* (%)	8361 (53.2%)	3681 (58.2%)	8456 (49.8%)	508 (50.4%)
Type 2 diabetes, *n* (%)	994 (6.5%)	900 (15.6%)	1431 (8.4%)	126 (12.5%)
Hypertension, *n* (%)	3248 (21.2%)	2666 (46.2%)	3275 (19.3%)	387 (38.4%)
Current smoker, *n* (%)	4015 (25.5%)	1771 (28.0%)	5498 (32.4%)	133 (13.2%)
COHORT COMPOSITION				
n (%) from MESA	1629 (10.3%)	1260 (19.9%)	538 (3.2%)	1007 (100.0%)
n (%) from CARDIA	1863 (11.8%)	1747 (27.6%)	0 (0.0%)	0 (0.0%)
n (%) from FOS	2393 (15.1%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
n (%) from ARIC	9926 (62.8%)	3314 (52.4%)	0 (0.0%)	0 (0.0%)
n (%) from CKB	0 (0.0%)	0 (0.0%)	16 455 (96.8%)	0 (0.0%)

Values are presented as mean (standard deviation), median (interquartile range), or n (percentage), as appropriate. Follow-up is observed time from baseline to first CHD event or censoring. Event rates equal CHD events divided by summed individual observed person-years, multiplied by 100. Percentages use available data. Sex was missing for 96 White participants from FOS; the White sex denominator is 15 715. Chinese-group event rates describe a sample dominated by outcome-dependent CKB selection and are not population incidence estimates.

Distribution of apoB, TG, Lp(a), and RW-apoB

The distributions of TG and Lp(a) differed markedly across ethnic groups, with higher Lp(a) levels in African American participants and higher TG levels in Chinese participants (*Figure 1*). ApoB distributions differed less substantially, and RW-apoB distributions showed considerable overlap despite different underlying lipid phenotypes.

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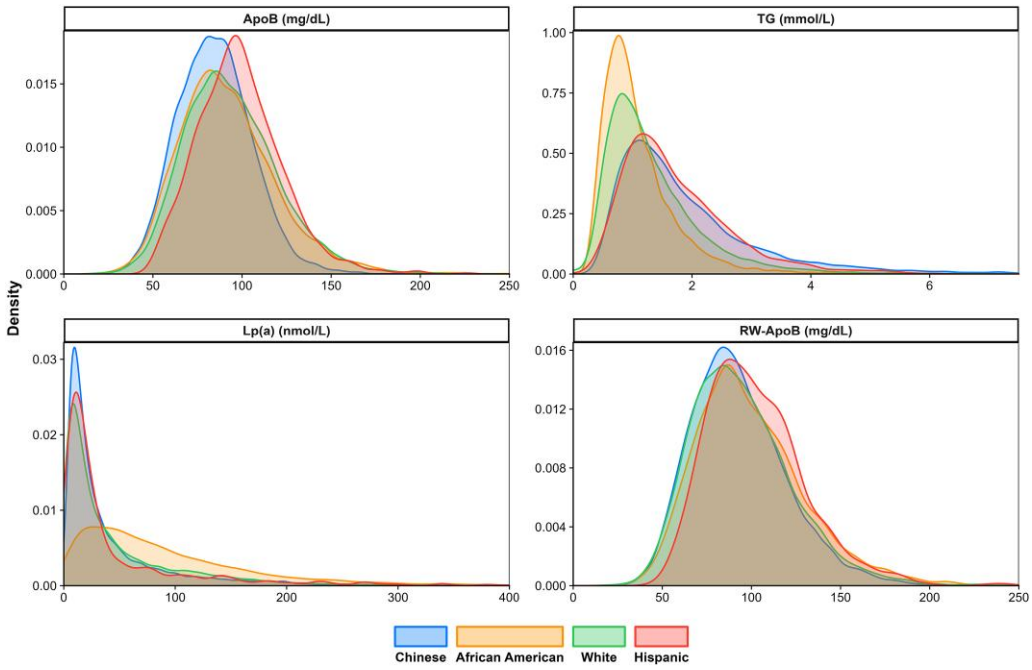


Figure 1 Distribution of apolipoprotein B, triglycerides, lipoprotein (a), and risk-weighted apolipoprotein B across ethnic groups. Kernel density plots display the full biomarker distributions, including spread, skewness, and overlap. Chinese participants are shown in blue, African American participants in orange, White participants in green, and Hispanic participants in red. Biomarker concentrations are shown on the *x*-axes in the stated units; the *y*-axis is probability density, and the area under each curve equals 1.

CKB participants were sampled in the non-fasting state, which may partly explain their higher TG levels. However, among fasting MESA participants, median TG were also higher in Chinese than in African American participants (1.4 vs. 1.0 mmol/L), despite lower body mass index and a lower prevalence of diabetes (23.8 vs. 30.0 kg/m^2 and 7% vs. 11%, respectively).

Mean RW-apoB and apoB were similar in White participants (95.0 and 94.0 mg/dL) and differed by 4.3 mg/dL in Hispanic participants (102.9 and 98.6 mg/dL). RW-apoB exceeded apoB by 7.9 mg/dL in African American participants (99.9 vs. 92.0 mg/dL) and by 8.5 mg/dL in Chinese participants (93.3 vs. 84.8 mg/dL). These differences corresponded predominantly to higher Lp(a) and TG, respectively. Median concentrations and percentile distributions are shown in *Table 1*.

Reclassification by RW-apoB

Cross-tabulation of ethnicity-specific quintiles of apoB and RW-apoB demonstrated substantial discordance across all ethnic groups (*Figure 2*). Within each group, participants were both up- and down-ranked when classified by RW-apoB rather than apoB, indicating differences between particle number and risk-weighted composition.

Characteristics of discordant individuals

Participants were categorized according to the difference between RW-apoB and apoB percentile ranks within each ethnic group (*Table 2*). Across all groups, upward re-ranking by RW-apoB was associated with higher TG and/or Lp(a). Both components contributed in White and Hispanic participants, Lp(a) contributed predominantly in African American participants, and TG contributed predominantly in Chinese participants. Participants in Q1 had relatively high apoB but lower RW-apoB, whereas those in Q5 had lower apoB but higher RW-apoB because of higher TG and/or Lp(a). The inverse apoB pattern reflects the definition of the re-ranking groups.

Associations of apoB and RW-apoB with coronary events

In mutually adjusted analyses (*Figure 3*), RW-apoB was associated with CHD conditional on apoB in the total pooled cohort (see Supplementary material online, *Figure S1*) and in all four ethnic sub-groups. HRs per 10 mg/dL were 1.11 (1.07 −1.14) in the total cohort, 1.11 (1.05–1.18) in White, 1.11 (1.08–1.15) in African American, 1.08 (1.05–1.11) in Chinese, and 1.10 (1.01–1.21) in Hispanic participants. Conditional apoB estimates were 1.01(1.00–1.03), 1.02 (1.00–1.05), 1.00 (0.97–1.04), 1.01 (0.96–1.07), and 0.96 (0.86–1.08), respectively; the White estimate was close to the conventional significance threshold (*P* rounded to 0.05 in *Figure 3*). Heterogeneity for RW-apoB was substantial in White participants (I^2 = 70.4%) and absent in the African American and Chinese pooled estimates (I^2 = 0%). Spearman correlations ranged from 0.79 to 0.91 and VIFs from 2.3 to 5.3 (see Supplementary material online, *Table S2*).

In separate single-marker models (*Figure 4*), C-indices for RW-apoB vs. apoB were 0.599 vs. 0.585 $P < 0.001$ in the total

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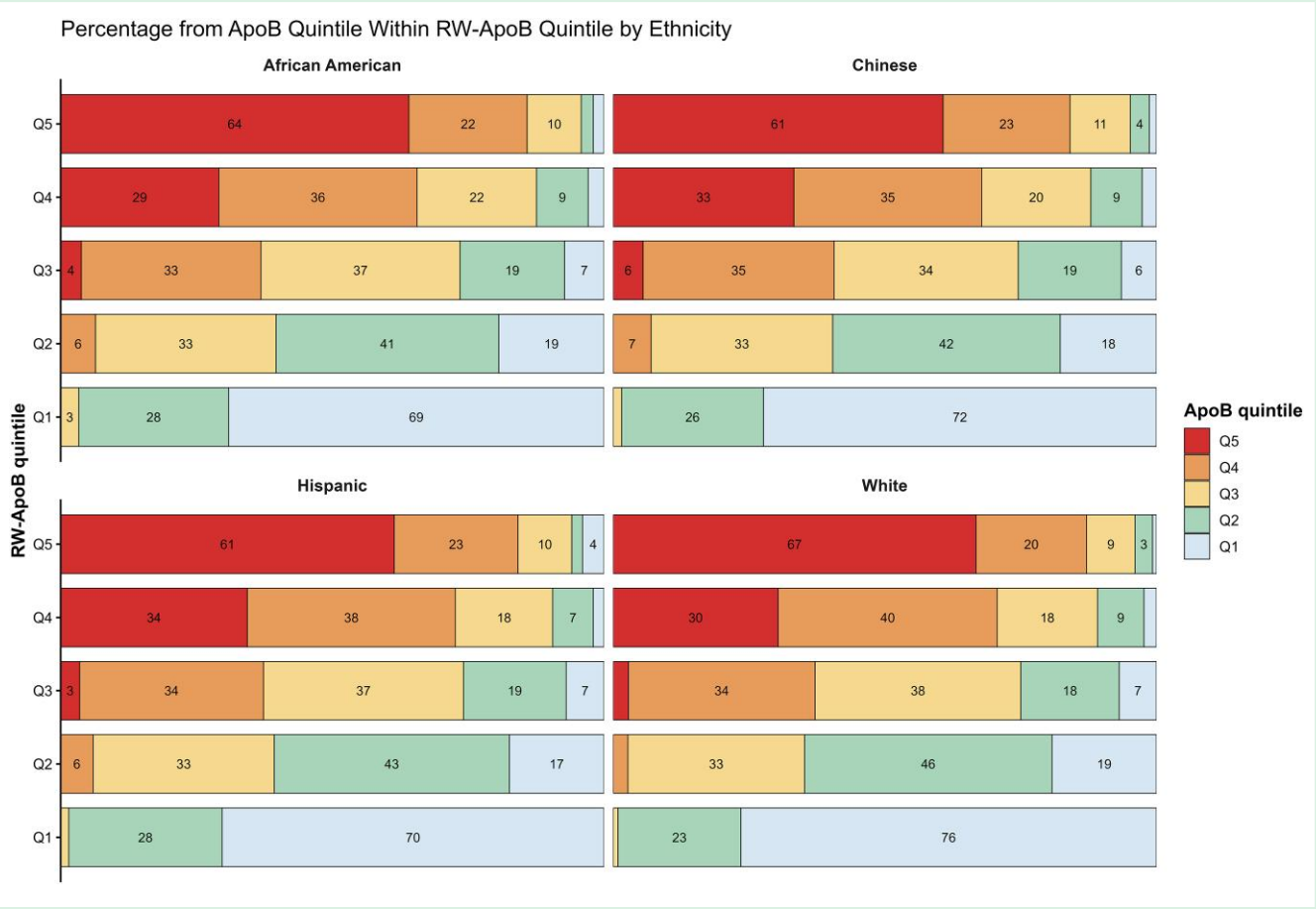


Figure 2 Discordance between apolipoprotein B and risk-weighted apolipoprotein B classifications across ethnic groups. Stacked bars show the distribution of apolipoprotein B quintiles within each ethnicity-specific risk-weighted apolipoprotein B quintile. Each bar represents 100% of participants in the corresponding risk-weighted apolipoprotein B quintile and illustrates the degree of concordance and discordance between the two measures.

pooled cohort (see supplementary material online, *Figure S2*), 0.630 vs. 0.617 in White participants ($P < 0.001$), 0.606 vs. 0.595 in African American participants ($P = 0.085$), 0.567 vs. 0.554 in Chinese participants ($P = 0.014$), and 0.540 vs. 0.535 in Hispanic participants ($P = 0.756$). Thus, discrimination was numerically higher in all four groups, with statistically significant improvements in the total pooled cohort and in the larger White and Chinese sub-groups. The absolute differences were modest. In White participants, non-HDL-C and RW-apoB had similar point estimates (0.628 and 0.630).

Sensitivity analyses

Age- and sex-adjusted HRs for RW-apoB conditional on apoB were 1.11 (1.07–1.15), 1.12 (1.08–1.16), 1.08 (1.05–1.11), and 1.12 (1.03–1.23) in White, African American, Chinese, and Hispanic participants, respectively (see Supplementary material online, *Table S5*). Corresponding per-SD estimates from models without age and sex adjustment were 1.34 (1.17–1.55), 1.41 (1.27–1.57), 1.22 (1.13–1.32), and 1.32 (1.02–1.71) (see Supplementary material online, *Table S6*).

Scaled Schoenfeld tests showed no statistically significant departures from proportional hazards in all models (global P = 0.075–0.933; Supplementary material online, *Table S3*). With follow-up truncated at 10 years, RW-apoB remained associated conditional on apoB in ARIC (HR 1.15, 1.11–1.18) and FOS (1.29, 1.16–1.44; Supplementary material online, *Table S4*). Retaining the 211 low-apoB MESA participants yielded HRs per 10 mg/dL of 1.09 (1.05–1.14) for RW-apoB and 0.95 (0.90–1.01) for apoB (see Supplementary material online, *Table S7*).

In hard-event sensitivity analyses limited to White and African American participants in ARIC and Chinese participants in CKB, RW-apoB remained associated after adjustment for apoB (all $P < 0.001$), whereas apoB was not associated after adjustment for RW-apoB ($P = 0.06$, $P = 0.87$ and $P = 0.34$, respectively). *Figure 3*.

In CKB, the mutually adjusted logistic OR was 1.08 (1.05–1.11; $P = 1.7 \times 10^{-6}$) for RW-apoB and 1.02 (0.98–1.07; $P = 0.243$) for apoB. Additional adjustment for age, age squared, sex, and region yielded ORs of 1.08 (1.04–1.11; $P = 1.12 \times 10^{-5}$) and 1.03 (0.99–1.08; $P = 0.182$), respectively. Single-marker AUCs were 0.575 (0.558–0.592) for RW-apoB and 0.561 (0.544–0.578) for apoB (difference 0.014, 0.004–0.025; paired DeLong $P = 0.008$; Supplementary material online, *Table S8*). These were unweighted analyses within the

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Table 2 Characteristics of participants with the greatest downward or upward re-ranking by risk-weighted apolipoprotein B relative to apolipoprotein B across ethnic groups

	Chinese		African American		White		Hispanic	
Characteristic	**Q1**	**Q5**	**Q1**	**Q5**	**Q1**	**Q5**	**Q1**	**Q5**
N	3399	3398	1265	1264	3163	3162	202	201
RW-apoB mg/dL, mean (SD)	87.2 (13.1)	108.0 (23.1)	89.0 (14.7)	112.6 (26.0)	87.6 (14.3)	107.5 (24.6)	94.7 (14.1)	116.7 (25.1)
apoB mg/dL, mean (SD)	94.6 (13.0)	77.6 (13.8)	99.6 (16.4)	79.9 (16.3)	102.2 (16.4)	84.3 (16.4)	106.5 (14.6)	89.4 (16.0)
(Ethnicity-specific RW-apoB percentile—apoB percentile), mean (SD)	−21.3 (5.7)	27.6 (13.7)	−21.1 (5.5)	27.7 (13.6)	−18.0 (4.8)	25.4 (13.3)	−20.0 (5.1)	27.3 (15.5)
Triglycerides mmol/L, mean (SD)	1.2 (0.4)	3.0 (1.7)	0.8 (0.3)	1.4 (1.3)	0.8 (0.3)	1.9 (1.4)	1.1 (0.4)	2.3 (1.2)
Lp(a) nmol/L, mean (SD)	17.6 (12.2)	74.7 (68.2)	29.6 (19.1)	174.5 (80.5)	13.2 (11.4)	108.5 (76.0)	13.9 (10.1)	112.8 (103.6)
LDL-C mg/dL, mean (SD)	109.5 (19.3)	81.1 (18.4)	127.8 (29.5)	123.1 (33.4)	132.0 (29.9)	125.9 (32.0)	133.9 (20.8)	113.6 (30.5)
HDL-C mg/dL, mean (SD)	51.5 (11.7)	44.0 (11.0)	54.2 (13.8)	54.8 (18.0)	54.1 (14.8)	48.8 (17.1)	51.2 (12.9)	44.0 (13.5)
Non-HDL-C mg/dL, mean (SD)	143.8 (23.3)	128.9 (27.2)	141.9 (31.3)	146.3 (35.6)	146.2 (34.9)	158.0 (36.4)	154.4 (23.8)	152.4 (29.9)
BMI kg/m^2, mean (SD)	23.5 (3.4)	23.9 (3.4)	30.4 (6.3)	29.9 (6.4)	26.0 (4.6)	27.5 (5.1)	29.1 (5.4)	29.2 (4.6)
Age, years, mean (SD)	57.7 (10.5)	56.4 (10.5)	41.0 (18.1)	52.3 (11.6)	44.3 (14.2)	53.1 (9.5)	62.0 (10.3)	60.3 (10.1)
Male, *n* (%)	1781 (52.4%)	1706 (50.2%)	591 (46.7%)	503 (39.8%)	1492 (47.8%)	1467 (46.5%)	102 (50.5%)	116 (57.7%)
Female, *n* (%)	1618 (47.6%)	1692 (49.8%)	674 (53.3%)	761 (60.2%)	1628 (52.2%)	1689 (53.5%)	100 (49.5%)	85 (42.3%)
Type 2 diabetes, *n* (%)	257 (7.6%)	332 (9.8%)	106 (9.8%)	220 (18.0%)	102 (3.4%)	285 (9.1%)	15 (7.4%)	34 (16.9%)
Hypertension, *n* (%)	621 (18.3%)	686 (20.2%)	378 (35.2%)	687 (56.1%)	368 (12.3%)	866 (27.7%)	76 (37.6%)	84 (41.8%)
Current smoker, *n* (%)	1083 (31.9%)	1136 (33.4%)	298 (23.6%)	372 (29.4%)	854 (27.4%)	750 (23.8%)	19 (9.4%)	30 (14.9%)

Values are presented as mean (standard deviation) or *n* (%). Participants were grouped according to quintiles of the ethnicity-specific difference between RW-apoB percentile and apoB percentile. Q1 indicates the greatest downward re-ranking by RW-apoB relative to apoB, and Q5 indicates the greatest upward re-ranking. These groups are not quintiles of apoB or RW-apoB concentration. Percentages for variables with missing data were calculated among participants with available information; consequently, category counts may not sum to the total group size.

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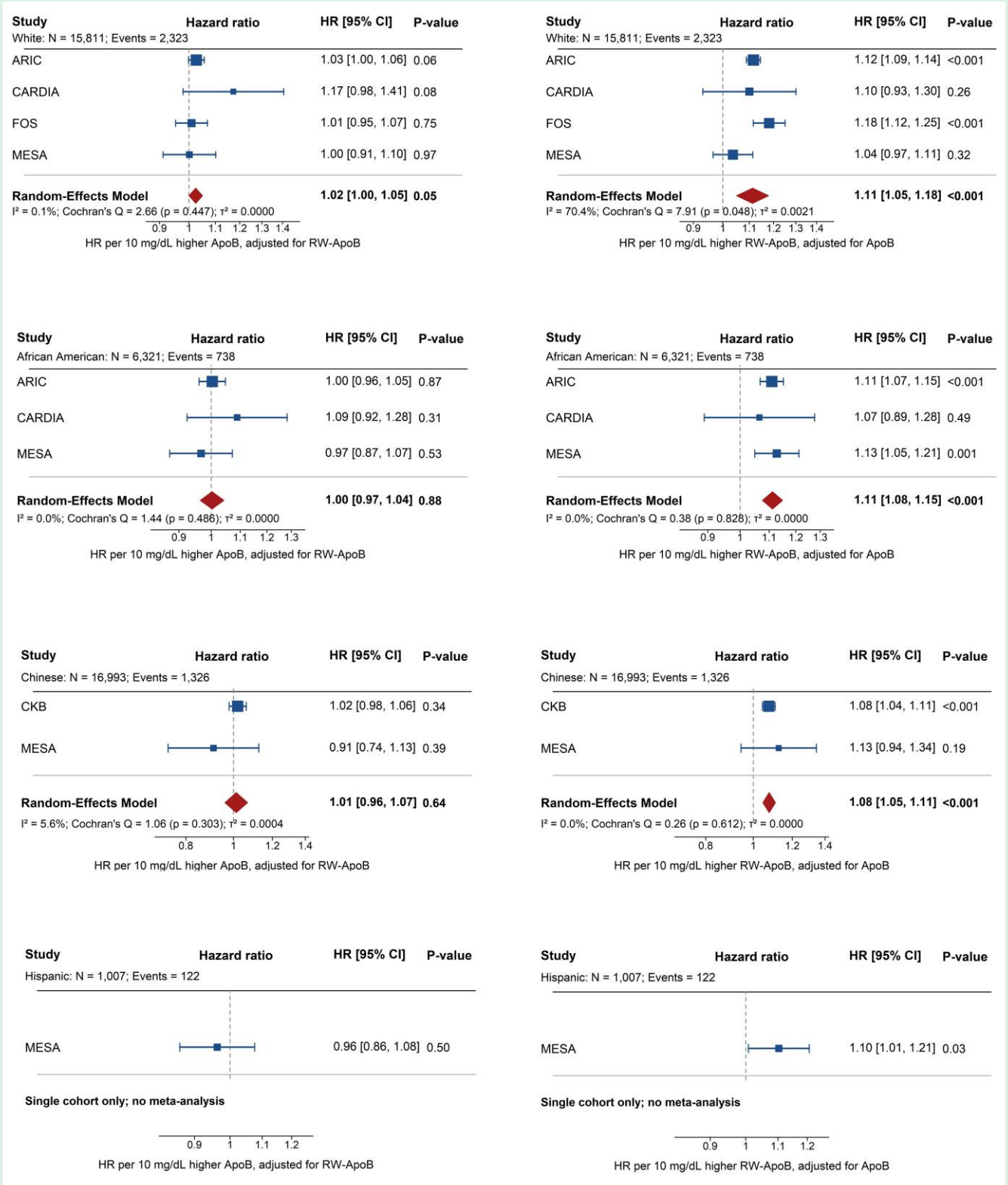


Figure 3 Cohort-specific and pooled associations of apolipoprotein B and risk-weighted apolipoprotein B with coronary heart disease across ethnic groups. Forest plots show mutually adjusted hazard ratios per 10 mg/dL with 95% CIs and *P*-values; total participant and event counts are given for each ethnic group. Estimates were pooled using random-effects meta-analysis (REML). Paired marker panels use the same axis scale within each ethnic group. The conditional coefficients do not represent two independent marker effects. China Kadoorie Biobank estimates are from an unweighted outcome-dependent sample.

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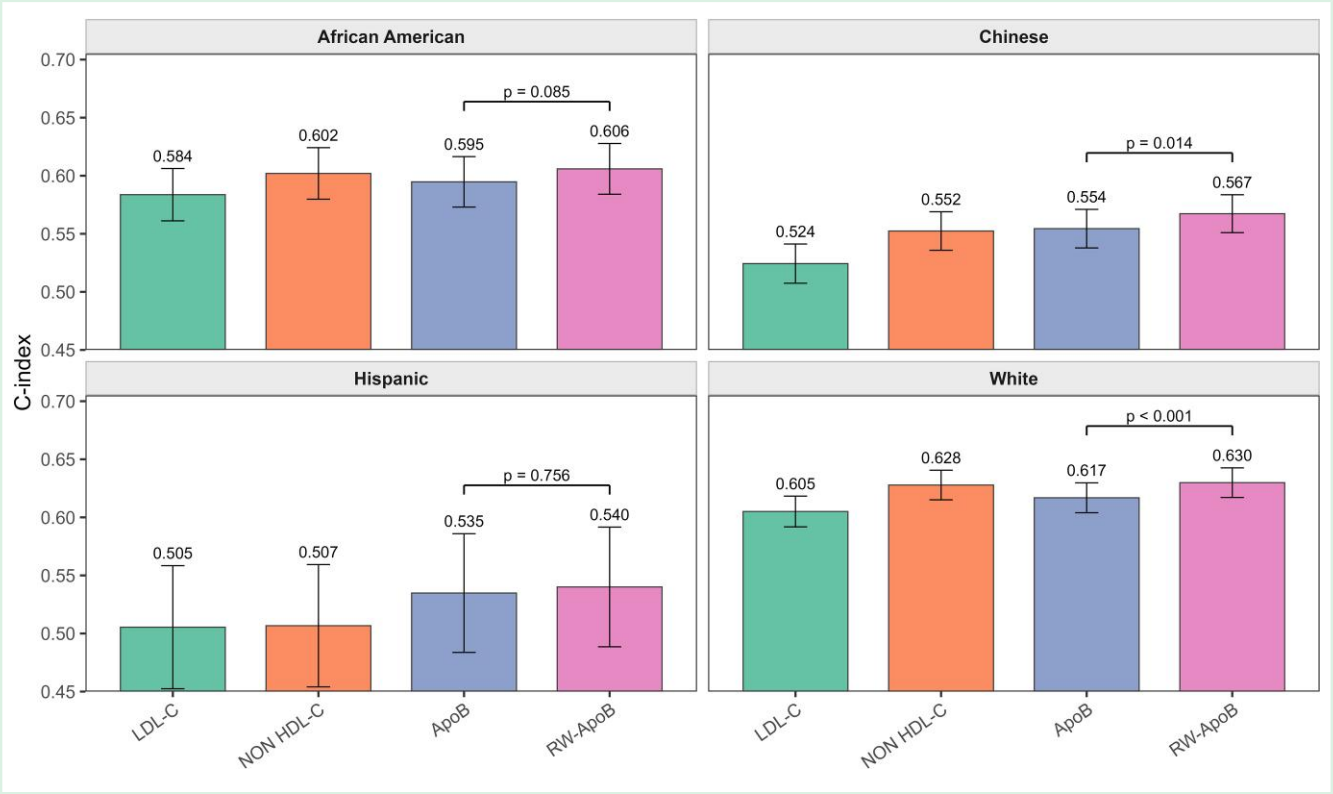


Figure 4 Discrimination of individual lipid measures for incident coronary heart disease across ethnic groups, stratified by cohort. Harrell's C-index and 95% confidence intervals are shown for low-density lipoprotein cholesterol, non-high-density lipoprotein cholesterol, apolipoprotein B, and risk-weighted apolipoprotein B from separate single-marker Cox models. *P*-values compare apolipoprotein B with risk-weighted apolipoprotein B using a paired Wald test based on the joint covariance of the concordance estimates. These analyses assess discrimination, not calibration or incremental performance beyond a clinical risk model. Chinese-group estimates predominantly reflect the outcome-dependent China Kadoorie Biobank sample.

selected CKB sample and did not correct for the original outcome-dependent sampling design.

Within White ARIC participants with apoB in the lowest tertile, event rates across increasing RW-apoB tertiles were 0.39, 0.40, and 0.93 per 100 person-years; corresponding rates in the middle apoB tertile were 0.36, 0.64, and 0.86 (see Supplementary material online, *Table S9*). The CKB sample showed a similar descriptive pattern (see Supplementary material online, *Table S10*). In African American MESA participants with apoB below the median, rates were 0.52 vs. 0.95 below vs. above the RW-apoB median (see Supplementary material online, *Table S11*). These unadjusted descriptions do not establish calibrated risk reclassification.

Discussion

In this multi-cohort study of 40 132 participants across four ethnic groups, RW-apoB captured differences in lipoprotein composition that were associated with CHD beyond measured apoB. Higher RW-apoB relative to apoB predominantly reflected higher Lp(a) in African American participants and higher TG in Chinese participants. RW-apoB had numerically higher discrimination than apoB in all four ethnic groups, with statistically significant improvements in the total pooled cohort, and in the larger White and Chinese groups.

These findings suggest that apoB alone may not fully capture the lipoprotein-associated component of CHD risk and that the extent and nature of this difference vary across populations. While apoB reflects the total number of atherogenic particles, it does not account for differences in particle composition.[3,8,9] RW-apoB incorporates this information by weighting the estimated contributions from TRL and Lp(a), which appear to confer greater risk per particle than LDL.[11]

A key observation is that similar average differences between RW-apoB and apoB in African American and Chinese participants arose from different underlying mechanisms. In African American participants, higher RW-apoB reflected higher Lp(a) concentrations, consistent with well-established ethnic differences in Lp(a) distribution.[12,13] In Chinese participants, higher RW-apoB reflected higher TG levels at comparable apoB concentrations, in line with prior reports of TG-enriched lipoprotein profiles in Chinese populations.[5,20] These findings suggest that the composition of apoB-containing lipoproteins differs systematically across ethnic groups, and that a single measure based on particle number alone may obscure these differences.

Consistent with this, individuals who were re-ranked upward by RW-apoB exhibited distinct phenotypes across ethnic

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groups. In White and Hispanic participants, reclassification reflected combined elevations in TG and Lp(a), whereas in African American participants it was primarily driven by Lp(a), and in Chinese participants by TG. Despite these differences, RW-apoB remained associated with CHD events after adjustment for apoB in all groups. This is consistent with prior work showing that TRLs and Lp(a) contribute independently to coronary risk beyond LDL-related particle burden.[6–9]

RW-apoB had a numerically higher C-index than apoB in the total pooled cohort and in every ethnic group, although the improvement was modest and statistically demonstrable only in the total pooled cohort and in the larger Chinese and White subgroups. In White participants, non-HDL-C and RW-apoB performed similarly. These findings support the concept that combining particle number and particle composition may provide additional lipoprotein-associated information, but they do not establish superiority to every conventional lipid marker in every population.

Association and discrimination measure different properties: RW-apoB may capture additional lipoprotein-associated information beyond apoB while producing only a small change in rank-based discrimination because the two measures are strongly correlated. The observed C-index range (approximately 0.51–0.63) is expected for models containing only one lipid marker and should not be compared with thresholds used for complete multivariable clinical prediction models. LDL-C, for example, had lower discrimination in these analyses despite being an established causal and clinically important risk factor. The C-index values were also similar to those in our previous European Heart Journal study.[11]

Mutual adjustment requires care because measured apoB is a component of RW-apoB. The association of RW-apoB after adjustment for apoB therefore reflects the information carried by its weighted TG and Lp(a) component at a given apoB concentration; it should not be interpreted as an independent effect of RW-apoB as a whole. Likewise, a weak conditional apoB coefficient does not call into question the association or causal relevance of apoB. The correlation between the two measures reduces precision and may make the coefficients less stable. Moreover, one coefficient being statistically significant while the other is not does not establish that their associations differ. The consistently positive conditional association for RW-apoB is nonetheless unlikely to be explained by multicollinearity alone. The separate, paired C-index comparisons address a different question: whether RW-apoB ranks participants by risk better than apoB.

The findings were broadly consistent with our previous UK Biobank study,[11] but they do not establish superiority over every conventional lipid marker. In White participants, the conditional apoB estimate was close to the significance threshold, while non-HDL-C and RW-apoB had similar C-indices. Differences in cohort composition, outcome definitions, follow-up, assays, and precision may contribute to variation between studies.

These findings may have implications for lipid assessment. LDL-C, non-HDL-C, and apoB can underestimate lipoprotein-associated risk when Lp(a) or TRLs are elevated.[1,3,11] Our results suggest that the lipid phenotype underlying such discordance differs across populations. RW-apoB may therefore provide a concise summary of particle number and composition, but its clinical utility and incremental value beyond established multivariable risk assessment require further study.

Several limitations should be acknowledged. Measurement protocols, fasting status, Lp(a) assays and conversions, and CHD outcome definitions differed across cohorts. In ARIC, historical measurements required conversion from the reported protein component to estimated total Lp(a) mass and subsequently to molar concentration. Although we followed the published ARIC protocol, fixed mass-to-molar conversions cannot fully account for variation in apo(a) isoform size and may introduce measurement error. TG were measured non-fasting in CKB but fasting in the U.S. cohorts, which may affect comparisons of absolute TG and RW-apoB concentrations. However, RW-apoB was developed using non-fasting UK Biobank samples, and fasting Chinese participants in MESA also had relatively high TG levels. Non-fasting sampling would not be expected to systematically favour RW-apoB over apoB in the outcome analyses; rather, any additional variability would be expected to attenuate associations.

Coronary event definitions varied across cohorts, ranging from MI and coronary death to broader composites that included angina and revascularization. Nevertheless, the findings were consistent in ARIC and CKB, where outcomes were restricted to hard coronary events. The RW-apoB coefficients were derived from published relative atherogenicity estimates and scaled in the predominantly White UK Biobank. Applying the unchanged equation tests transportability, but optimal weights may differ across populations.

Ethnicity and cohort were partly overlapping: Hispanic participants were represented only by MESA, and Chinese participants were predominantly from CKB. Random-effects meta-analysis allowed cohort-specific associations to vary and incorporated between-cohort heterogeneity into pooled uncertainty, but it cannot fully separate ethnicity from cohort or remove measurement error, residual confounding, and differences in outcome ascertainment. The estimates should therefore be interpreted as associations within the available cohort composition rather than as causal differences between ethnic groups. Age- and sex-adjusted sensitivity analyses were consistent with the primary RW-apoB associations, but the analyses do not constitute comprehensive clinical prediction models.

CKB contributed 41% of participants and 28% of events and accounted for almost all Chinese participants. Its Lp(a) measurements originated from a nested case–control sample selected for CHD and stroke. Because the unweighted Cox models do not account for this outcome-dependent sampling, the magnitude of the CKB hazard ratios should be interpreted cautiously. However, CKB-specific logistic regression yielded directionally consistent results: RW-apoB remained associated with CHD after adjustment for apoB, whereas apoB was not associated with CHD after adjustment for RW-apoB. This supports the presence of additional lipoprotein-associated information captured by the TG/Lp(a)-weighted component of RW-apoB within the selected CKB sample, although it does not validate the precise magnitude of the Cox HR. The paired ROC analysis also showed modest but significantly higher discrimination for RW-apoB than apoB within the analysed CKB sample.

Single baseline measurements over prolonged follow-up are susceptible to regression dilution and may not capture subsequent changes in treatment or exposure. Although the 10-year

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analyses in ARIC and FOS were directionally consistent with the primary analyses, no correction for regression dilution was applied. However, such exposure variability would generally be expected to attenuate rather than strengthen the observed associations. Moreover, tests based on scaled Schoenfeld residuals showed no evidence of violation of the proportional hazards assumption. As in our previous analysis, we excluded MESA participants with apoB <50 mg/dL because some may have been receiving lipid-lowering treatment that was not recorded. Including them gave similar results in MESA overall, although we did not repeat this comparison within each ethnic group.

In conclusion, RW-apoB captured lipoprotein-associated CHD information beyond measured apoB across the studied ethnic groups, despite different lipid phenotypes. Discrimination was higher in the total pooled cohort and numerically higher in all groups, with statistically significant improvements in the larger White and Chinese groups. However, calibration, clinical utility, or incremental predictive value beyond comprehensive risk models remain to be established. Nevertheless, RW-apoB may serve as a more accurate summary metric of lipoprotein-associated risk compared to apoB.

Supplementary material

Supplementary material is available at European Journal of Preventive Cardiology.

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ChatGPT (OpenAI) was used to assist with editorial revision, consistency checking, and document preparation. No participant-level data were supplied to the tool.

Author contributions

Michaela B Rehman (Conceptualization, Formal analysis, Methodology, Writing—original draft, Writing—review & editing [equal], Data curation [lead]), and Elias Björnson (Conceptualization, Formal analysis, Methodology, Writing—original draft, Writing—review & editing [equal])

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Data availability

Data from the FOS, MESA, ARIC and CARDIA studies are available from the respective study repositories and National Institutes of Health subject to institutional approvals and data use agreements. Data from the CKB is available upon application to the CKB Access Management System.

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